

A Brief History of Computing

This short document traces the history of computing across several eras, from mechanical calculators to the global Internet. Each page covers a distinct period.

Early Mechanical Calculators

In the seventeenth century Blaise Pascal built a mechanical adding machine, and Gottfried Leibniz later designed a device that could also multiply. In the nineteenth century Charles Babbage designed the Analytical Engine, a general-purpose mechanical computer that was never fully built in his lifetime. Ada Lovelace, working from Babbage's designs, wrote what is often considered the first algorithm intended to be carried out by a machine, and is remembered as the first computer programmer.

The Theoretical Foundations

In 1936 Alan Turing introduced an abstract model of computation now called the Turing machine, which formalized the idea of an algorithm and defined the limits of what can be computed. Around the same time Alonzo Church developed the lambda calculus, an equivalent model.

In 1937 Claude Shannon showed that Boolean algebra could describe the behavior of electrical switching circuits, linking logic to hardware. A decade later Shannon founded information theory, giving a precise, mathematical meaning to the quantity of information measured in bits. Together these ideas form the theoretical bedrock of computer science.

The First Electronic Computers

During the 1940s the first large-scale electronic computers were built. ENIAC, completed in 1945, used thousands of vacuum tubes and could be reprogrammed only by rewiring it physically.

The decisive idea of this era was the stored-program concept, in which a computer's instructions are held in the same memory as its data. This design is commonly called the von Neumann architecture, after a 1945 report by John von Neumann, and almost every general-purpose computer since has followed it.

The Transistor and the Integrated Circuit

In 1947 researchers at Bell Labs invented the transistor, a compact, reliable replacement for the fragile vacuum tube. In 1958 the integrated circuit packed many transistors onto a single chip.

In 1965 Gordon Moore observed that the number of transistors on a chip was doubling roughly every two years, an empirical trend later known as Moore's law. In 1971 Intel released the 4004, the first commercial microprocessor, placing a complete processing unit on one chip and opening the way to personal computers.

Networking and the Internet

In 1969 ARPANET connected its first few university computers, demonstrating packet switching over long distances. In the 1970s Vinton Cerf and Robert Kahn designed the TCP/IP protocol suite, which let independent networks interconnect and became the technical foundation of the Internet.

In 1989 Tim Berners-Lee proposed the World Wide Web at CERN, and by 1991 the first web pages were online. The Web combined hypertext, the HTTP protocol, and the URL addressing scheme to make the Internet broadly accessible, setting the stage for the connected world of today.